

Problem Set – Introduction to Functions.

1. Allow the user to repeatedly enter a quantity and price. Prompt the user on whether they want

to do the program (Yes or No). Use a function to compute the total (quantity times price).

The

function should be passed the quantity and price and then return the total. In the function, provide a 10% discount if the total is over \$10,000.00. Display quantity, price and total.

Sum

and display the extended price.

Input	Process	Output
qty, price	Function: CompExtPrice(qty, unitprice)	
	ExtPrice = qty * unitprice	newExtPrice (total після знижки)
	If ExtPrice > 10000	
	DiscAmt = ExtPrice * 0.10	
	Else DiscAmt = 0	
	newExtPrice = ExtPrice – DiscAmt	
	return newExtPrice	
qty, price, response	Main	
	totalExtPrice = 0	quantity, price, total (для кожного рядка)
	Ask: “Do you want to do this program (Yes or No)” → response	totalExtPrice (сума всіх total)
	While response == "Yes"	
	Input qty, price	
	newExtPrice = CompExtPrice(qty, price)	
	Display qty, price, newExtPrice	
	totalExtPrice = totalExtPrice + newExtPrice	

	Ask again response (Yes or No)	
	After loop: Display totalExtPrice	

2. Enter players last name, number of hits and at bats at the keyboard. Prompt the user on whether they want to do the program (Yes or No). Use a function to compute batting average. Pass the hits and at bats to the function. The function should return batting average. Display last name and batting average. Give a count of the number of players entered.

Input	Process	Output
hits, at_bats	Function: CompBatAvg(hits, at_bats)	
	If at_bats > 0	battingAvg
	battingAvg = hits / at_bats	
	Else battingAvg = 0	
	return battingAvg	
lastName, hits, at_bats, response	Main	
	playerCount = 0	lastName, battingAvg (для кожного гравця)
	Ask: "Do you want to do this program (Yes or No)" → response	playerCount (кількість гравців)
	While response == "Yes"	
	Input lastName, hits, at_bats	
	avg = CompBatAvg(hits, at_bats)	
	Display lastName, avg	
	playerCount = playerCount + 1	

	Ask again response	
	After loop: Display playerCount	

3. Enter the destination city, miles travelled and gallons used for a trip. Prompt the user on whether they want to do the program (Yes or No). Use a function to compute miles per gallon and cost of gas. Pass miles travelled and gallons used to the function. The function should return miles per gallon and compute gas cost to be gallons times 3.00. Count the number of entries made (number of trips) Display destination city, miles, mpg and gas cost. At end display the number of entries made, total miles travelled for all trips and total gas cost of all trips.

Input	Process	Output
miles, gallons	Function: CompTrip(miles, gallons)	
	mpg = miles / gallons	mpg
	gasCost = gallons * 3.00	gasCost
	return mpg, gasCost	
city, miles, gallons, response	Main	
	tripCount = 0	For each trip: city, miles, mpg, gasCost
	totalMiles = 0	At end: tripCount, totalMiles, totalGasCost
	totalGasCost = 0	

	Ask: "Do you want to do this program (Yes or No)" → response	
	While response == "Yes"	
	Input city, miles, gallons	
	mpg, gasCost = CompTrip(miles, gallons)	
	Display city, miles, mpg, gasCost	
	tripCount = tripCount + 1	
	totalMiles = totalMiles + miles	
	totalGasCost = totalGasCost + gasCost	
	Ask again response	
	After loop: Display tripCount, totalMiles, totalGasCost	

4. Allow the employee to enter last name, job code and hours worked. Prompt the user on whether they want to do the program (Yes or No). Use a function to determine the pay rate. Pass to this function the job code and it should return rate of pay and gross pay. Use Job code L is \$25/hr, A is \$30/hr and J is \$50/hr for respective pay rates. Compute gross pay. Give time and a half for overtime. Display last name, hours, pay rate and gross pay. Sum and display total of all gross pay.

Input	Process	Output
jobCode, hours	Function: CompPay(jobCode, hours)	
	If jobCode == 'L' → rate = 25	rate, grossPay

	Else if jobCode == 'A' → rate = 30	
	Else if jobCode == 'J' → rate = 50	
	If hours > 40	
	regular = 40 * rate	
	otHours = hours - 40	
	otPay = otHours * rate * 1.5	
	grossPay = regular + otPay	
	Else grossPay = hours * rate	
	return rate, grossPay	
lastName, jobCode, hours, response	Main	
	totalGross = 0	For each employee: lastName, hours, rate, grossPay
	While response == "Yes"	totalGross (сума всіх grossPay)

Input lastName, jobCode, hours
 rate, gross = CompPay(jobCode,
 hours)
 Display lastName, hours, rate,
 gross
 totalGross = totalGross + gross
 Ask again response
 After loop: Display totalGross

5. Allow the user to enter student last name, credit hours and district code. Prompt the user on whether they want to do the program (Yes or No). Use a function to compute tuition owed.

Charge In district (code of I) \$250 per credit hour. Out of district (code of O) is \$550 per credit

hour. The function should receive credit hours and district code and return tuition owed. Display

student name and tuition owed. Sum and display total of all tuition owed.

Input	Process	Output
credits, distCode	Function: CompTuition(credits, distCode)	
	If distCode == 'I' → rate = 250	tuition
	Else if distCode == 'O' → rate = 550	
	Else rate = 0	
	tuition = credits * rate	
	return tuition	
lastName, credits, distCode, response	Main	
	totalTuition = 0	For each student: lastName, tuition
	Ask: “Do you want to do this program (Yes or No)” → response	totalTuition (сума усієї оплати)
	While response == "Yes"	
	Input lastName, credits, distCode	
	tuition = CompTuition(credits, distCode)	

	Display lastName, tuition	
	totalTuition = totalTuition + tuition	
	Ask again response	
	After loop: Display totalTuition	